Thomas Thuan F. Le Nguyen

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Research Interests

Data Science, Machine Learning, Computer Vision, Agentic AI, and Modeling and Simulation.

Education

Florida International University — Miami, FL
M.S. in Data Science and Artificial Intelligence — Expected Spring 2027
GPA: 3.92

California State University San Marcos — San Marcos, CA
B.S. in Computer Science, Minor in Mathematics — Fall 2023
GPA: 3.34

Research Experience

Researcher —— Compound Flooding — Sep 2026–Present
Florida International University — Miami, FL
Research Supervisor: Dr. Jason Liu

- Developing Model Context Protocols (MCPs) to enable AI agents to perform and optimize task-specific workflows.
- Investigating the effects and interactions of compound flooding events through environmental data collection, processing, and analysis.

Software Developer — MOSAIC (Mobile Sensing and Analytics) Lab — Oct 2025–Present
Florida International University — Miami, FL
Lab Director: Dr. Christian Poellabauer
Faculty Collaborator: Dr. Shanna Burke

- Design, develop, and test a mobile application for collecting participant-reported data on sleep, treatment, and daily well-being as part of a cancer survivorship research project funded in part by the Florida Cancer Innovation Fund.
- Develop and maintain a web-based administration platform for managing research studies and creating and configuring surveys and activities.
- Design and maintain database infrastructure supporting research study configuration, survey and activity management, and participant data management.
- Implement and maintain software functionality based on requirements provided by the research team and collaborators.
- Software developed is also used by Baptist Health South Florida in clinical surgery-related studies.

Undergraduate Researcher — Summer Scholars Program — Jun 2021–Aug 2021
California State University San Marcos — San Marcos, CA
Research Supervisor: Dr. Ali Ahmadinia

- Conducted Genentech-sponsored undergraduate research focused on computer vision for low-cost embedded systems.
- Developed and tested YOLOv4 TFLite object detection models using Google Colab.
- Evaluated models on a Coral Development Board, benchmarking inference speed (FPS) and mean Average Precision (mAP).

- Presented research findings at the Summer Scholars Undergraduate Research Program Final Research Showcase.

Presentations

Le Nguyen, Thomas Thuan F. "Object Detection Using Low Cost Embedded Systems"
Summer Scholars Undergraduate Research Program Final Research Showcase, California State University San Marcos, 2021. [Poster]

Selected Data Science & AI Projects

Electric Vehicle Ownership Prediction

- Analyzed and visualized electric vehicle adoption across Washington State counties using geospatial and demographic data.
- Examined relationships between EV adoption and factors including education, income, and political affiliation.
- Explored adoption trends over time and compared patterns across geographic regions.
- Developed visualizations to communicate findings and highlight regional trends.

Perceptron and Adaline Models

- Implemented and evaluated Perceptron and Adaline neural networks from scratch for vowel classification.
- Created a custom training set and three independent test sets to evaluate model performance.
- Compared model accuracy using quantitative results, graphs, and tables.

Payphone — AI-Enabled Point-of-Sale System

- Led a team of eight in developing a point-of-sale system incorporating artificial intelligence and machine learning for product identification and automated subtotal calculation.
- Integrated a database for inventory management across multiple store locations.

Teaching Experience

Instructional Student Assistant Jan 2022–Dec 2023
CSIS Department, California State University San Marcos San Marcos, CA

- Supported more than 500 students across courses in introductory programming, data structures, assembly language, digital circuits, and operating systems.
- Clarified advanced computer science concepts, graded assignments and examinations, and collaborated with course instructors.
- Communicated recurring student questions and challenges to instructors to improve course support.

STEM Instructor May 2024–Aug 2024
CodeREV Kids Los Angeles, CA

- Taught more than 300 students in computer programming, engineering, and 3D modeling.
- Conducted school enrichment sessions focused on circuitry and coding.

Lead Coding Instructor Apr 2019–May 2024
Code Ninjas Encinitas, CA

- Taught programming to more than 2,000 students using JavaScript, Lua, C#, and Java.
- Led multi-day coding camps and managed classes of 20–30 students.
- Conducted school enrichment sessions covering robotics, drones, and computer programming.

FLL Jr. Mentor Aug 2017–Jun 2018

La Clase Magica Solana Beach, CA

- Mentored kids from underserved communities for the FIRST Lego League Jr. competition.
- Helped plan and guided students to build a water system out of Legos.
- Taught the importance of clean water and illustrated the different water systems and cleaning services.

Industry Experience

Software Engineer Intern Jun 2023–Aug 2023
Hunter Industries San Marcos, CA

- Developed a minimum viable product for visualizing precise customer information.
- Developed methods for retrieving previously inaccessible data from customer controllers.
- Applied Test Driven Development (TDD), Agile, and Scrum methodologies.

Disney College Program — Merchandise Aug 2024–May 2025
Walt Disney World Orlando, FL

- Operated point-of-sale systems and assisted guests with purchases.
- Managed inventory, restocked merchandise, and maintained organized retail environments.

Technical Skills

Programming: Python, C++, Java, JavaScript, C#, Lua
Machine Learning: Computer Vision, Deep Learning, Natural Language Processing, Transformers
Frameworks: PyTorch, TensorFlow, Scikit-learn
Data Science: Data Analysis, Data Visualization, Model Evaluation
Tools: Linux, Git, Docker, Google Colab